

Course Name: Computer Network and Internet Protocol

Assignment - Week 0 (Jan 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total mark: 10 X 1 = 10

QUESTION 1:

Which scheduler in the operating system is responsible for swapping processes from the waiting queue to and from secondary memory?

- a) Medium Term scheduler
- b) Long Term scheduler
- c) Short Term scheduler
- d) None of above

Correct Answer: (a)

Detailed Solution: Medium term scheduler is responsible for swapping process from waiting and ready queue to and from secondary memory.

QUESTION 2

What is the function of the umask command in Linux?

- a) To alter file ownership
- b) To adjust file permissions
- c) To establish default file permission mask
- d) To reveal hidden files

Correct Answer: (c)

Detailed Solution : The umask command establishes the default permission mask for newly generated files and folders. It specifies the permissions that are not automatically conferred.

QUESTION 3

What is the "response time" of a process in CPU scheduling? "-" indicates the minus sign.

- a) Initial response time - Arrival time
- b) Completion time - Burst duration

- c) Turnaround time - Burst duration
- d) First response time - Waiting duration

Correct Answer: (a)

Detailed solution: Response time is the duration between a process's arrival in the ready queue to its first allocation of CPU for execution. Response Time = Time of Initial Response – Arrival Time.

QUESTION 4

Identify the octal expression of the file permission -rw-r-xr--.

- a) 654
- b) 564
- c) 444
- d) 555

Correct Answer: (a)

Detailed solution: read(r) represented by 4, write(w) represented by 2 and execute(x) represented by 1 so, it will be 654.

QUESTION 5

In Linux, which command is used to print system date and time?

- a) datetime
- b) sysdatetime
- c) time
- d) date

Correct Answer: (d)

Detailed solution: date command will show system date, time and time zone.



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QUESTION 6

In a system with a single-level page table, the TLB access time is 20 nanoseconds, and the main memory access time is 80 nanoseconds. If the TLB hit ratio is 95%, what is the effective memory access time (EMAT) in nanoseconds?

- a. 94
- b. 96
- c. 100
- d. 98

Correct Answer: (b)

Detailed Solution: $EMAT = \text{Hit (TLB access time + Main Memory access time)} + \text{Miss (TLB access time + Page table access time + Main Memory access time)}$.

Given: TLB access time = 20 ns

Main Memory access time = 80 ns

TLB hit ratio = 95%

$$\begin{aligned} EMAT &= 0.95(20 + 80) + 0.05(20 + 80 + 80) \\ &= 0.95(100) + 0.05(180) \\ &= 95 + 9 \\ &= 96 \text{ nanoseconds.} \end{aligned}$$

QUESTION 7

What occurs if signal() is called on a semaphore without a matching wait()?

- a) The semaphore value reaches zero.
- b) The semaphore value is increased, and no error transpires.
- c) The process will await the semaphore value to reach 0.
- d) An error has occurred.

Correct Answer: (b)

Detailed Solution : The use of signal() increases the semaphore's value, regardless of whether a matching wait() has been executed. This is legitimate, and no errors are present.

QUESTION 8

Identify the full form of HTTP?

- a) Hypertext Transfer Protocol
- b) Hypertext Transfer Package
- c) Hyper Transfer Text Package
- d) Hyper Transfer Text Practice

Correct Answer: (a)

QUESTION 9

What mathematical operation denotes the time complexity of Shortest Job First (SJF) scheduling for a queue of n processes?

- a) $O(n)$
- b) $O(n \log n)$
- c) $O(n^2)$
- d) $O(1)$

Correct Answer: (b)

Detailed Solution : The Shortest Job First (SJF) scheduling algorithm necessitates the arrangement of processes based on their burst durations, resulting in a temporal complexity of $O(n \log n)$. This renders it computationally efficient, if not linear.

QUESTION 10:

Which of the following statements is/are true for an inverted page table in case of memory management strategies of the operating system?

- (i) Searching time is more than conventional page table
 - (ii) Saves memory in RAM than conventional page table
- a) Only (i)
 - b) Only (ii)
 - c) Both (i) and (ii)
 - d) Neither (i) nor (ii)

Correct Answer: (c)